

```

from collections import deque

maze = [
    ['S', '.', '.', '#'],
    ['#', '.', '#', '.'],
    ['.', '.', '.', '.'],
    ['#', '#', '.', 'G']
]

rows = len(maze)
cols = len(maze[0])

for i in range(rows):
    for j in range(cols):
        if maze[i][j] == 'S':
            start = (i, j)
        if maze[i][j] == 'G':
            goal = (i, j)

queue = deque([start])
visited = {start}
parent = {}

directions = [(-1,0), (1,0), (0,-1), (0,1)]

while queue:
    x, y = queue.popleft()

    if (x, y) == goal:
        break

    for dx, dy in directions:
        nx, ny = x + dx, y + dy

        if (0 <= nx < rows and
            0 <= ny < cols and
            maze[nx][ny] != '#' and
            (nx, ny) not in visited):

            visited.add((nx, ny))
            parent[(nx, ny)] = (x, y)
            queue.append((nx, ny))

path = []
node = goal

while node != start:
    path.append(node)
    node = parent[node]

path.append(start)
path.reverse()

print("Shortest Path:")
print(path)
print("Steps:", len(path)-1)

```